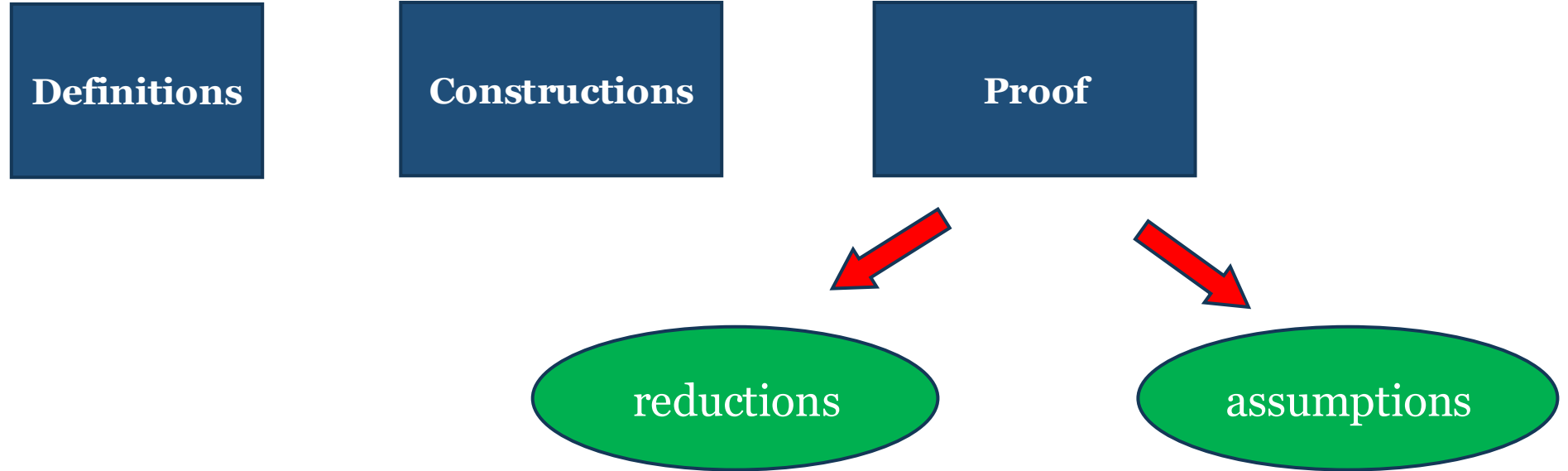
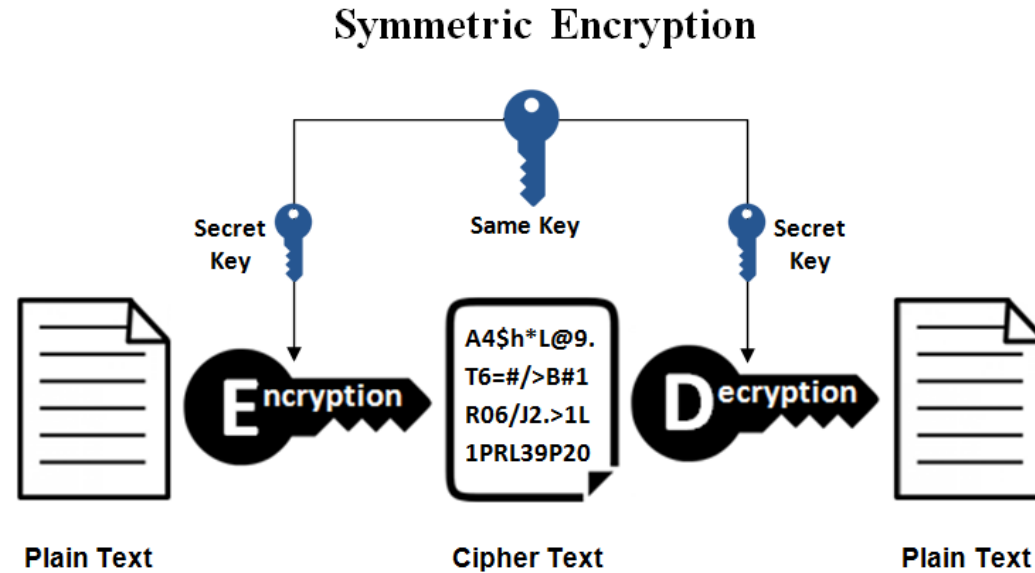


ONE-TIME PAD

Modern Cryptography



Symmetric-key Encryption



Definitions for encryption

- Algorithms:

K : secret key, M : message, C : ciphertext

- $K = \mathbf{Gen}()$
- $C = \mathbf{Enc}(k, m)$
- $M = \mathbf{Dec}(k, c)$

Notation

- $\mathcal{K}$ (key space) – set of all possible keys
 - $\mathcal{C}$ (ciphertext space) – set of all possible ciphertexts
 - $\mathcal{M}$ (message space) - set of all possible messages
-

Definitions for encryption

- Algorithms:

K : secret key, M : message, C : ciphertext

- $K = \text{Gen}()$
- $C = \text{Enc}(k, m)$
- $M = \text{Dec}(k, c)$

- Correctness:** for every message $m \in \mathcal{M}$, any secret key $k \in \mathcal{K}$



Probability

- *Event*: a particular occurrence in some experiment
 - $\Pr[E]$: probability of event E
- Conditional probability: probability that one event occurs, *given that* some other event occurred
 - $\Pr[A \mid B] = \Pr[A \text{ and } B] / \Pr[B]$
- Two r.v.'s X, Y are *independent* if
for all x, y : $\Pr[X = x \mid Y = y] = \Pr[X = x]$

Probability

- Law of total probability: say $E_1, \dots, E_n$ are a *partition* of all possibilities. Then for any A :

$$\Pr[A] = \sum_i \Pr[A \text{ and } E_i] = \sum_i \Pr[A \mid E_i] \cdot \Pr[E_i]$$

Definitions for encryption

- Algorithms:

K : secret key, M : message, C : ciphertext

- $K = \text{Gen}()$
- $C = \text{Enc}(k, m)$
- $M = \text{Dec}(k, c)$

- **Perfect secrecy:** for any distribution over the message space $\mathcal{M}$, for every message $m \in \mathcal{M}$ and every ciphertext $c \in \mathcal{C}$,

$$\Pr[M = m \mid C = c] = \Pr[M = m]$$

An equivalent definition

- Algorithms:

K : secret key, M : message, C : ciphertext

- $K = \text{Gen}()$
- $C = \text{Enc}(k, m)$
- $M = \text{Dec}(k, c)$

- **Perfect secrecy:** for any distribution over the message space $\mathcal{M}$, for every message $m \in \mathcal{M}$ and every ciphertext $c \in \mathcal{C}$,

$$\Pr[C = c | M = m] = \Pr[C = c]$$

Proof

- Bayes's theorem: $\Pr[A \mid B] \cdot \Pr[B] = \Pr[B \mid A] \cdot \Pr[A]$
- $\Pr[M = m \mid C = c] \Pr[C = c] = \Pr[C = c \mid M = m] \Pr[M = m]$
- If $\Pr[M = m \mid C = c] = \Pr[M = m]$,
then $\Pr[C = c \mid M = m] = \Pr[C = c]$

Theorem 1(Shannon 1949) : If (Enc, Dec) is perfectly secure, then $|\mathcal{M}| \leq |\mathcal{K}|$

Proposition 1: For every two messages $m_0, m_1 \in \mathcal{M}$ and any ciphertext $c \in \mathcal{C}$,

$$Pr[Enc_k(m_0) = c] = Pr[Enc_k(m_1) = c]$$

where the probability is over the random choice of k . An encryption scheme is perfectly secure if and only if the equation above holds

- **Theorem 1**: If (Enc, Dec) is perfectly secure, then $|\mathcal{M}| \leq |\mathcal{K}|$
- If $|\mathcal{M}| > |\mathcal{K}|$, fix M_0 , and any key K_0 and $C_0 = \text{Enc}(K_0, M_0)$
- $\Pr_k [C_0 = \text{Enc}(K, M_0)] > 0$
- $S = \{\text{Dec}(C_0, k) : k \in \mathcal{K}\}$, $|S| \leq |\mathcal{K}| < |\mathcal{M}|$
- $\exists M_1, \forall k, \text{Enc}(k, m_1) \neq C_0$
- $\Pr_k [C_0 = \text{Enc}(K, M_1)] = 0$

Proposition 1: For every two messages $m_0, m_1 \in \mathcal{M}$ and any ciphertext $c \in \mathcal{C}$,

$$\Pr_k[Enc_k(m_0) = c] = \Pr_k[Enc_k(m_1) = c]$$

where the probability is over the random choice of k . An encryption scheme is perfectly secure if and only if the equation above holds

Proof.

One-time pad

- Key length the same as message length

XOR

- $A \oplus B$

input		output
0	0	0
0	1	1
1	0	1
1	1	0

Equivalent to addition mod 2

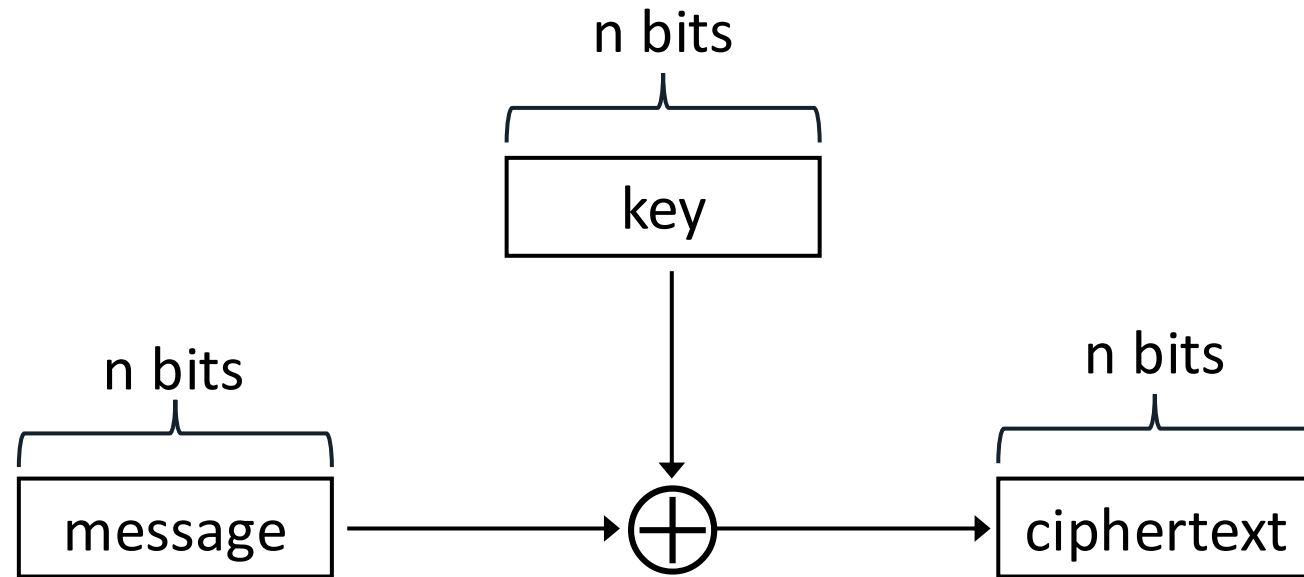
One-time pad

- Let $\mathcal{M} = \{0,1\}^n$
- $\text{Gen}()$: choose a uniform key $k \in \{0,1\}^n$
- $\text{Enc}_k(m)$: $c = k \oplus m$
- $\text{Dec}_k(c)$: $m = k \oplus c$

- Correctness:

$$\begin{aligned}\text{Dec}_k(\text{Enc}_k(m)) &= k \oplus (k \oplus m) \\ &= (k \oplus k) \oplus m = m\end{aligned}$$

One-time pad



Theorem

- Theorem 1: One-time pad encryption is perfectly secure.

Perfect secrecy: for any distribution over the message space $\mathcal{M}$, for every message $m \in \mathcal{M}$ and every ciphertext $c \in \mathcal{C}$,

$$\Pr[C = c | M = m] = \Pr[C = c]$$

Proof

- $\Pr[C = c | M = m] = \Pr[k \oplus m = c | M = m]$
 $= \Pr[k = m \oplus c | M = m] = 2^{-n}$

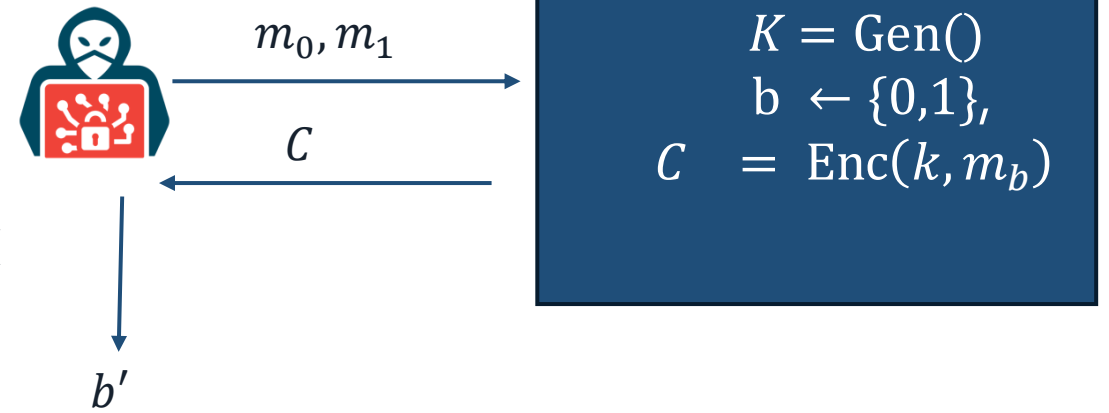
Proof

- $$\begin{aligned}\Pr[C = c] &= \sum_{m \in \mathcal{M}} \Pr[C = c | M = m] \Pr[M = m] \\ &= \sum_{m \in \mathcal{M}} 2^{-n} \Pr[M = m] \\ &= 2^{-n} \sum_{m \in \mathcal{M}} \Pr[M = m] \\ &= 2^{-n}\end{aligned}$$

Another def: perfect indistinguishability

Experiment:

1. Adversary A chooses any two messages $m_0, m_1 \in \mathcal{M}$
2. Generate a key k using $\text{Gen}()$, pick a random bit $b \in \{0,1\}$, send attacker $c = \text{Enc}(k, m_b)$
3. Attacker guesses b'
4. Experiment outputs succeed if $b = b'$



Perfectly indistinguishable: for every adversary A

$$\Pr[\text{Experiment outputs } 1] = 1/2$$

Perfect indistinguishability

- *Theorem 2: An encryption scheme is perfectly secret if and only if it is perfectly indistinguishable.*

Proposition 1: *For every two messages $m_0, m_1 \in \mathcal{M}$ and any ciphertext $c \in \mathcal{C}$,*

$$\Pr_k[Enc_k(m_0) = c] = \Pr_k[Enc_k(m_1) = c]$$

where the probability is over the random choice of k . An encryption scheme is perfectly secure if and only if the equation above holds

Proof.

$$\exists m_0, m_1, c^*. Pr_k[Enc_k(m_0) = c^*] > Pr_k[Enc_k(m_1) = c^*]$$



m_0, m_1

c

```
K = Gen()
b ← {0,1},
C = Enc(k, m_b)
```

If $c = c^*$, output $b' \leftarrow 0$

Otherwise, output random bit $b' \leftarrow \{0, 1\}$

Using the same key twice?

- $m_1 \oplus m_2$ is information about m_1, m_2
- Is this significant?
 - No longer perfectly secret!
 - $m_1 \oplus m_2$ reveals where m_1, m_2 differ
 - Frequency analysis